

Game-Guided Tactical Corridor Learning for Autonomous Racing Overtakes

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ABSTRACT

Overtaking planning and decision-making are central to performance in autonomous racing. Limited maneuvering space on narrow closed circuits makes high-speed overtaking difficult and increases the risk of failed passes and collisions. To address this problem, we formulate interactive overtaking as robust tactical feasible-set selection and develop a game-guided decision framework that distills Stackelberg response-set reasoning into persistent tactical corridors for constrained motion planning and closed-loop execution. First, a robust Stackelberg response-set teacher provides action-prior and action-conditioned robust-value supervision to a Truncated Quantile Critics (TQC) policy that learns passing-side intention, corridor parameters, and speed demand. Second, a hysteretic state machine preserves the selected passing side, while a deterministic corridor carver combines the learned corridor parameters with the evolving track and interaction geometry to construct a time-varying Frenet corridor. Finally, a corridor-constrained sequential second-order cone programming (SOCP) planner and a speed profiler generate a continuous path-speed reference, which is tracked using reference-conditioned contouring-error nonlinear model predictive control (RC-NMPC) to produce closed-loop vehicle commands. Hardware-in-the-loop experiments were conducted using an EAV24 formula race-car simulator. Across 1000 laps, the method achieved an overtaking success rate of 94.7% and a mean overtaking time of 4.82 s. Compared with three baselines using the same execution stack, the success rate was 31.2–66.2 percentage points higher and the mean overtaking time was 3.01–6.41 s shorter; no collisions or track-boundary violations were observed.

1. Introduction

Autonomous racing provides a demanding testbed for integrated perception, decision-making, planning, and control under extreme operating conditions. Competitions such as the Indy Autonomous Challenge and the Abu Dhabi Autonomous Racing League (A2RL) have extended this testbed to full-scale head-to-head racing near the limits of vehicle handling^{1,2}. In these races, the ability to complete overtaking maneuvers safely and efficiently directly affects competitive performance. Overtaking is especially difficult on narrow closed circuits, where high speed and limited lateral space shorten the passing window and increase collision risk. A viable overtaking decision must retain its passing-side intent across repeated replanning and leave sufficient maneuvering freedom for downstream path generation and vehicle control. The central challenge is therefore to translate interactive tactical reasoning into a geometrically explicit and temporally persistent feasible region throughout the overtaking maneuver.

1.1. Related work

Research on overtaking in autonomous racing has developed along four main directions: hierarchical maneuver planning, explicit game-theoretic planning, learning-based decision-making, and hybrid learning-optimization architectures. In hierarchical architectures, a tactical layer selects among following, defending, and passing behaviors, while a lower layer converts the selected behavior into a path and speed profile. Finite-state and candidate-trajectory planners provide a structured basis for real-time computation and allow racing rules, collision checks, and vehicle constraints to be handled by dedicated modules^{3,4}. This modular structure is also adopted in full-scale head-to-head racing systems, where opponent prediction and local planning operate under tight real-time constraints^{2,5}. Within such a hierarchy, Thakkar et al. formulated the high-level decision as a discrete racing game that generates target waypoints for either a multi-agent reinforcement-learning controller

or a linear-quadratic Nash-game controller⁶. These approaches provide interpretable tactical logic and a practical route to real-time implementation. Their tactical outputs, however, are commonly represented by maneuver states, target waypoints, or selected trajectories and provide only limited continuous parameterization of the spatial extent available to downstream planning.

Explicit game-theoretic planners model the coupling between competitors more directly. Non-cooperative dynamic games jointly evaluate progress, collision avoidance, blocking, and overtaking, producing behaviors such as shielding, faking, and opportunistic passing that are difficult to obtain from independent trajectory optimization^{7,8}. Iterative linear-quadratic approximations reduce the computational cost of solving nonlinear multi-player games, although their local solution quality remains sensitive to the interaction model and initialization⁹. Hierarchical Stackelberg formulations separate the leader's maneuver choice from the follower's response, while payoff learning adapts the interaction objective from observed behavior^{10,11}. Level- K reasoning offers a bounded model of opponent rationality; trajectory mixing can further hedge against changes in the inferred reasoning level¹². Constrained-potential and near-potential formulations instead exploit common structure in the players' objectives to reduce the online game-solving burden^{13,14}. Collectively, these methods make strategic coupling explicit. Despite computational approximations, repeatedly solving the game online remains demanding under the real-time constraints of autonomous racing.

Learning-based methods replace repeated online search with fixed-cost policy inference and acquire interaction patterns directly from competitive experience. Curriculum learning has produced overtaking policies by gradually increasing opponent difficulty, while large-scale deep reinforcement learning has demonstrated high-performance racing and human-like tactical behavior in simulation^{15–17}. Comparisons with optimal control show that reinforcement learning can exploit complex racing dynamics effectively, although its behavior depends strongly on the training distribution and

reward design¹⁸. Other learning-based designs condition policies on nominal trajectories or infer actions from local LiDAR observations^{19,20}. Data-driven predictive planners use learned opponent forecasts to generate overtaking trajectories²¹. Game-informed deep reinforcement learning couples the policy reward to another vehicle's action, whereas dynamic-graph and multi-agent formulations encode pairwise relations or coordinated behavior among several vehicles^{22–24}. Human feedback provides another source of policy guidance for stabilizing learned driving strategies²⁵. These methods improve interaction awareness. Direct-control policies and single-trajectory outputs, however, leave limited freedom for constraint-aware replanning as the interaction evolves during overtaking.

Hybrid learning–optimization architectures therefore expose an intermediate decision to a model-based planner. Reiter et al. used reinforcement learning to parameterize the objective of a nonlinear predictive planner, confining exploration to trajectories accepted by the optimizer²⁶. In road-traffic applications, learned tactical layers have been used to anticipate disruptive lane changes and to coordinate merging sequences subsequently realized by optimal control^{27,28}. Most closely related to corridor-based planner interfaces, Yang et al. used MARL-generated cooperative reference trajectories to construct spatio-temporal corridors for downstream optimization²⁹. Toschi et al. constructed alternative drivable tunnels from the predicted occupancies of other vehicles, selected a tunnel according to the current interaction, and optimized a trajectory within it³⁰. Across these approaches, intermediate outputs parameterize the planning objective, provide a guiding trajectory, or select among separately generated corridors; the corridor parameters themselves are not learned as tactical outputs.

In summary, despite substantial progress, maintaining a consistent overtaking intent while preserving sufficient geometric freedom for constrained replanning remains challenging throughout a high-speed overtaking maneuver. This motivates a decision framework that represents this intent as a persistent tactical feasible region and communicates game-supervised tactical decisions to the planner through side-locked Frenet bounds and speed guidance.

1.2. Contributions and paper organization

To address this challenge, this study formulates interactive overtaking as robust tactical feasible-set selection and develops a game-guided tactical-corridor decision framework. The framework converts robust Stackelberg response-set reasoning into a persistent tactical feasible region and speed guidance for constrained planning and closed-loop control. Specifically, the main contributions are as follows:

1. **Robust Stackelberg Response-Set Supervision:** We develop a robust Stackelberg response-set teacher that evaluates constraint-admissible tactical candidates against near-best follower responses. The teacher then derives a response-set-optimal action prior and action-conditioned robust-value labels, thereby supervising a Truncated Quantile Critics (TQC) actor-critic to learn robust overtaking decisions.
2. **Persistent Tactical Feasible-Region Interface:** We introduce a persistent tactical feasible-region interface that converts TQC outputs into a side-locked, time-varying Frenet corridor with speed guidance. A hysteretic state machine and deterministic opponent-footprint carving adapt the feasible region to evolving track and interaction geometry, enabling continuous support for path planning via sequential second-order cone programming (SOCP) and reference-conditioned

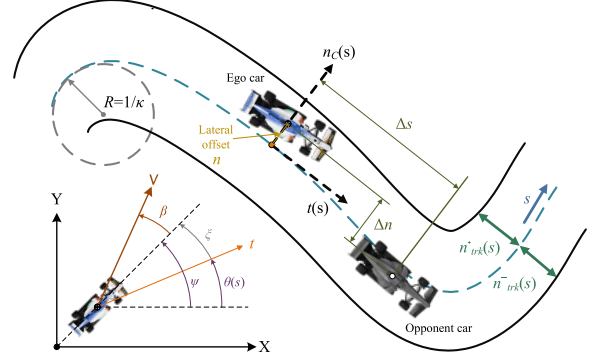


Fig. 1 Vehicle motion and relative geometry in the Frenet frame.

contouring-error nonlinear model predictive control (RC-NMPC) throughout overtaking.

3. **System Validation under High-Speed Conditions:** We deploy the complete decision-planning-control stack on the target onboard computer and evaluate it using an EAV24 hardware-in-the-loop platform under high-speed head-to-head racing conditions. Matched baseline comparisons and ablation studies are also conducted. The results demonstrate a high overtaking success rate and collision-free closed-loop operation, highlighting the framework's potential for practical deployment in autonomous racing.

The rest of this article is organized as follows. Section 2 formulates the problem and vehicle-track model. Sections 3 and 4 present the tactical-corridor framework and its planning-control realization, respectively. Section 5 reports the experimental results and discussion, and Section 6 concludes the paper.

2. Problem description and modeling

Figure 1 shows the two-vehicle overtaking problem considered in this study, in which the ego race car must overtake a non-cooperative opponent on a closed track. The following models define the vehicle and interaction quantities used by the tactical policy and executor. Both vehicles are represented by identical rectangular footprints centered at their state reference points, with length $L_j = 5.2$ m and width $W_j = 1.9$ m, $j \in \{e, o\}$.

For vehicle $i \in \{e, o\}$, (X_i, Y_i, ψ_i) denotes the inertial pose and $(v_{x,i}, v_{y,i}, r_i)$ the body-frame velocity and yaw rate, with $V_i = \sqrt{v_{x,i}^2 + v_{y,i}^2}$ and $\beta_i = \text{atan2}(v_{y,i}, v_{x,i})$. Let $\mathcal{W} = \{\text{fl}, \text{fr}, \text{rl}, \text{rr}\}$, and let $\mathbf{r}_q = [x_q, y_q]^\top$ and δ_q denote the body-frame position and steering angle of wheel q . Wheel-level quantities are vehicle specific, with the vehicle index omitted for clarity. The front wheels share the commanded steering angle and the rear wheels are unsteered.

The planar four-wheel dynamics are

$$\begin{bmatrix} \dot{X}_i \\ \dot{Y}_i \\ \dot{\psi}_i \\ \dot{v}_{x,i} \\ \dot{v}_{y,i} \\ \dot{r}_i \end{bmatrix} = \begin{bmatrix} v_{x,i} \cos \psi_i - v_{y,i} \sin \psi_i \\ v_{x,i} \sin \psi_i + v_{y,i} \cos \psi_i \\ r_i \\ \frac{1}{m} \sum_{q \in \mathcal{W}} (F_{x,q} \cos \delta_q - F_{y,q} \sin \delta_q) + r_i v_{y,i} \\ -\frac{F_{\text{drag}}(V_i)}{m} \\ \frac{1}{m} \sum_{q \in \mathcal{W}} (F_{x,q} \sin \delta_q + F_{y,q} \cos \delta_q) - r_i v_{x,i} \\ \frac{1}{I_z} \sum_{q \in \mathcal{W}} [x_q (F_{x,q} \sin \delta_q + F_{y,q} \cos \delta_q) \\ - y_q (F_{x,q} \cos \delta_q - F_{y,q} \sin \delta_q)] \end{bmatrix}. \quad (1)$$

Here m and I_z are the vehicle mass and yaw inertia, and F_{drag} is the aerodynamic drag force. The identified pure-slip tire forces are represented by

$$\begin{aligned} F_{d,q}(\chi_{d,q}, F_{z,q}) &= D_d(F_{z,q}) \sin(C_d \arctan(B_d \chi_{d,q})), \\ D_d(F_{z,q}) &= F_{z,q} \left[\mu_d + s_d \frac{F_{z,q} - F_{z,0}}{F_{z,0}} \right], \\ B_d &= \frac{\tan(\pi/(2C_d))}{q_{d,\text{pk}}}. \end{aligned} \quad (2)$$

Here $\chi_{x,q} = \lambda_q$ and $\chi_{y,q} = \alpha_q$. The resultant wheel forces determine the path-tangential and path-normal accelerations a_T and a_N . Following³¹, their admissible combinations are represented by the speed-interpolated envelope

$$\mathcal{A}_{\text{tire}}(V) = \{(a_T, a_N) : \Phi_{\text{tire}}(a_T, a_N, V) \leq 1\}. \quad (3)$$

Here Φ_{tire} is the normalized utilization returned by the interpolated map, with $\Phi_{\text{tire}} = 1$ defining the envelope boundary. The station-wise traction and braking magnitudes used in path planning are denoted by $a_{T,k}^+$ and $b_{T,k}^-$, respectively.

Let $\mathbf{r}_c(s)$ denote the track centerline parameterized by arc length $s \in [0, L_{\text{trk}}]$, with tangent angle $\theta(s)$ and curvature $\kappa(s) = d\theta/ds$, and let $\mathbf{t}(s)$ and $\mathbf{n}_c(s)$ be its unit tangent and normal. With $\mathbf{p}_i = [X_i, Y_i]^T$, the Frenet coordinates satisfy $\mathbf{p}_i = \mathbf{r}_c(s_i) + n_i \mathbf{n}_c(s_i)$ and $\xi_i = \psi_i - \theta(s_i)$. Defining the path-aligned velocity components as $v_{s,i} = V_i \cos(\xi_i + \beta_i)$ and $v_{n,i} = V_i \sin(\xi_i + \beta_i)$ gives

$$\begin{bmatrix} \dot{s}_i \\ \dot{n}_i \\ \dot{\xi}_i \end{bmatrix} = \begin{bmatrix} v_{s,i}/(1 - \kappa(s_i)n_i) \\ v_{n,i} \\ r_i - \kappa(s_i)v_{s,i}/(1 - \kappa(s_i)n_i) \end{bmatrix}. \quad (4)$$

The lateral coordinate is positive to the left, with $n_{\text{trk}}^-(s) \leq n \leq n_{\text{trk}}^+(s)$. Centerline projection gives the relative ego-opponent state

$$\begin{aligned} \Delta s &= \text{wrap}_{L_{\text{trk}}}(s_o - s_e), \\ \Delta n &= n_o - n_e, \quad \Delta v = v_{s,e} - v_{s,o}. \end{aligned} \quad (5)$$

Here $\text{wrap}_{L_{\text{trk}}}(\cdot)$ returns the signed arc-length difference in $[-L_{\text{trk}}/2, L_{\text{trk}}/2]$; $\Delta s > 0$ denotes an opponent ahead and $\Delta v > 0$ denotes the ego closing. The first sampled instant satisfying $\Delta s < 0$ is termed the *Pass* event throughout this paper.

For path generation, the same geometry is written in spatial form. Here s is the independent variable, $\mathbf{z}(s) = [n(s), \xi(s)]^T$ is the geometric state, and $\omega(s)$ is the planned-path curvature:

$$\mathbf{z}'(s) = \mathbf{f}_s(\mathbf{z}, \omega; s) = \begin{bmatrix} (1 - \kappa n) \tan \xi \\ (1 - \kappa n) \omega / \cos \xi - \kappa \end{bmatrix}. \quad (6)$$

3. Game-Supervised Tactical Feasible-Region Framework for Interactive Overtaking

The proposed tactical layer represents interactive overtaking decisions as persistent feasible regions for constrained motion planning. A robust Stackelberg response-set teacher evaluates constraint-admissible tactical candidates against near-best follower responses and generates a response-set-optimal action prior together with action-conditioned robust-value labels. A projected-action TQC actor-critic distills this supervision into passing-side intention, corridor parameters, and speed demand. A hysteretic state machine preserves the selected passing side, while a deterministic opponent-footprint carver converts the learned corridor parameters and evolving interaction geometry into time-varying Frenet corridors.

3.1. Robust Stackelberg Response-Set Supervision

At each tactical update, the encoder combines the two vehicle states, local track preview, current mode, side lock, and execution health into $\tilde{\mathbf{o}}_t$. The actor produces

$$\mathbf{a}^{\text{raw}} = [L^{\text{cap}}, R^{\text{cap}}, d_{\text{safe}}, V_{\text{cap}}^{\text{req}}, g_{\text{chase}}, g_{\text{ot}}, g_{\text{abort}}, b_{\text{lat}}, w_{\text{safe}}, q_{\text{side}}]^T. \quad (7)$$

where the components specify the left and right lateral caps, footprint clearance, requested speed cap, three mode-transition distances, defensive lateral offset and width, and passing-side intention. Together, they parameterize a planner-compatible tactical feasible region with speed guidance. The Euclidean projection $\Pi_{\mathcal{A}}$ maps the raw action into the admissible parameter domain by enforcing the component bounds in Table A.1, the ordered transition thresholds $g_{\text{ot}} + \Delta_g \leq g_{\text{chase}} \leq g_{\text{abort}} - \Delta_g$, and the minimum corridor width. This produces the bound-projected action $\mathbf{a}^{\text{B}}$, which is subsequently processed by the tire-admissibility projection Π_{tire} to obtain the final tactical action $\mathbf{a}^{\text{F}}$.

The offline teacher forms deterministic leader and follower banks by combining the three side codes with a Sobol design over the remaining action components. Every candidate is passed through the same bound projection, corridor carver, and tire screen used online; duplicate or infeasible candidates are discarded. Let $\mathbf{s}_t^G = (\tilde{\mathbf{o}}_t, \chi_t^G)$ denote the full rollout state available to the offline teacher, where χ_t^G collects the internal publisher, executor, and one-step reward memory required to evaluate a coupled candidate preview. The deployed policy receives only the 51-dimensional observation $\tilde{\mathbf{o}}_t$; the additional rollout variables χ_t^G are therefore treated as latent variables when the full-state teacher labels are transferred to the observation-conditioned policy. Let $\mathcal{A}_i^G(\mathbf{s}_t^G)$ denote the candidate bank for role i . Each candidate pair induces the coupled executable preview

$$(\mathcal{R}_e, \mathcal{R}_o) = \mathcal{P}_{N_{\text{sta}}}(\mathbf{s}_t^G; \mathbf{a}_e^{\text{F}}, \mathbf{a}_o^{\text{F}}), \quad (8)$$

where $\mathcal{P}_{N_{\text{sta}}}$ recursively applies the stateful publisher in Algorithm 1 and the common execution stack. The candidate pair remains fixed over the preview while both vehicle and hybrid states evolve jointly; both utilities use the corresponding role-relative features of this preview.

Let $Z_g = \sum_{k=0}^{N_{\text{sta}}-1} \gamma_g^k$ and $c_w = \sum_{j=1}^{10} [-w_j]_+$. The normalized role-relative utility is

$$U_i(\mathbf{s}_t^G, \mathbf{a}_e^{\text{F}}, \mathbf{a}_o^{\text{F}}) = \frac{1}{Z_g \|\mathbf{w}\|_1} \sum_{k=0}^{N_{\text{sta}}-1} \gamma_g^k [\mathbf{w}^T \boldsymbol{\phi}_{i,k}(\mathcal{R}_e, \mathcal{R}_o) + c_w], \quad (9)$$

where $\phi_i \in [0, 1]^{10}$ contains the role-relative reward features in Appendix A. Hence $U_i \in [0, 1]$, and ε_{BR} directly specifies the admissible follower-utility gap. For a fixed ego candidate, let $U_o^{\max}(s_t^G, \mathbf{a}_e^F)$ denote the largest follower utility over $\mathcal{A}_o^G(s_t^G)$. The near-best response set is

$$\mathcal{R}_o^{\varepsilon}(s_t^G, \mathbf{a}_e^F) = \{\mathbf{a}_o^F \in \mathcal{A}_o^G(s_t^G) : U_o(s_t^G, \mathbf{a}_e^F, \mathbf{a}_o^F) \geq U_o^{\max}(s_t^G, \mathbf{a}_e^F) - \varepsilon_{\text{BR}}\}. \quad (10a)$$

The teacher evaluates each candidate against its least favorable retained response and selects the strongest robust action:

$$\begin{aligned} \mathcal{V}_g(s_t^G, \mathbf{a}_e^F) &= \min_{\mathbf{a}_o^F \in \mathcal{R}_o^{\varepsilon}(s_t^G, \mathbf{a}_e^F)} U_e(s_t^G, \mathbf{a}_e^F, \mathbf{a}_o^F), \\ \mathbf{a}_{\text{th}}^*(s_t^G) &\in \arg \max_{\mathbf{a}_e^F \in \mathcal{A}_e^G(s_t^G)} \mathcal{V}_g(s_t^G, \mathbf{a}_e^F). \end{aligned} \quad (10b)$$

Here $\varepsilon_{\text{BR}} = 0.05$. Every evaluated ego candidate contributes the matched sample $(\bar{\mathbf{o}}_n, \mathbf{a}_{g,n}^F, v_{g,n}, \mathbf{a}_{\text{th},n}^*)$ to $\mathcal{D}_g$, where $v_{g,n} = \mathcal{V}_g(s_n^G, \mathbf{a}_{g,n}^F)$.

Before publication, the requested speed cap is screened against the speed-dependent tire envelope on the provisional corridor midpoint. For each trial $v \in \mathcal{V}_s(V_{\text{cap}}^{\text{req}})$, define the envelope-unclipped kinetic-energy template

$$\lambda_k = \frac{\sum_{j=0}^{k-1} \Delta \ell_j}{\sum_{j=0}^{N_{\text{sta}}-2} \Delta \ell_j}, \quad \widehat{V}_k(v) = \sqrt{V_e^2 + \lambda_k(v^2 - V_e^2)}. \quad (11)$$

Using the midpoint curvature $\widehat{\kappa}_k$, the induced acceleration demand and peak envelope utilization are

$$\begin{aligned} \widehat{a}_{T,k}(v) &= \frac{\widehat{V}_{k+1}^2(v) - \widehat{V}_k^2(v)}{2\Delta \ell_k} + \frac{F_{\text{drag}}(\widehat{V}_k(v))}{m}, \\ \widehat{a}_{N,k}(v) &= \widehat{\kappa}_k \widehat{V}_k^2(v), \\ \widehat{\rho}(v) &= \max_{0 \leq k \leq N_{\text{sta}}-2} \Phi_{\text{tire}}(\widehat{a}_{T,k}(v), \widehat{a}_{N,k}(v), \widehat{V}_k(v)). \end{aligned} \quad (12a)$$

The projection retains the largest trial cap within the envelope:

$$\begin{aligned} V_{\text{cap}}^F(\mathbf{a}) &= \max\{v \in \mathcal{V}_s(V_{\text{cap}}^{\text{req}}) : \widehat{\rho}(v) \leq 1\}, \\ \Pi_{\text{tire}}(\mathbf{a}; \bar{\mathbf{o}}) &= \mathbf{a}[V_{\text{cap}}^F(\mathbf{a})]. \end{aligned} \quad (12b)$$

Here V_e is the current ego speed, $\mathcal{V}_s$ is the uniformly sampled finite grid from $V_{\min}$ to $V_{\text{cap}}^{\text{req}}$, and $\mathbf{a}[v]$ changes only the cap component. If the grid contains no admissible value, Π_{tire} returns the current-state HOLD/braking action specified in Algorithm 1.

3.2. Game-Supervised Projected-Action TQC

The real-time policy maps the evolving duel geometry to the tactical feasible set at a fixed computational cost. Its 51-dimensional observation concatenates 27 duel-state features with two 12-dimensional track summaries centered at the ego and opponent. The duel features contain normalized Frenet states and accelerations, relative progress, lateral gap and closing rate, time to contact, closing-speed code, side lock, and a nine-state race-mode code. Each track summary reports mean curvature, peak absolute curvature, and minimum width over four consecutive bins.

The environment reward is

$$\begin{aligned} r_t &= \mathbf{w}^T \phi_t, \\ \phi_t &= [\phi_{\text{ot}}, \phi_{\text{ovb}}, \phi_{\text{col}}, \phi_{\text{lat}}, \phi_{\text{crush}}, \phi_{\text{app}}, \\ &\quad \phi_{\text{sm}}, \phi_{\text{gs}}, \phi_{\text{sbs}}, \phi_{\text{wide}}]^T. \end{aligned} \quad (13)$$

Appendix A gives the normalized definitions and weights. The same role-swapped features define Eq. (9), aligning the teacher

and online reward at the level of overtaking progress, clearance, smoothness, and usable corridor width.

Projected-action TQC. Truncated Quantile Critics (TQC)³² discard the largest target atoms before the quantile-Huber backup. Environment replay $\mathcal{D}$ stores the final tactical action $\mathbf{a}_t^F$. For a raw target-policy sample, the projected action and truncated target are

$$\begin{aligned} \mathbf{a}_{t+1}^F &= \Pi_{\text{tire}}(\Pi_{\mathcal{A}}(\mathbf{a}_{t+1}^{\text{raw}}); \bar{\mathbf{o}}_{t+1}), \\ \ell_{t+1} &= \log \pi_{\theta}(\mathbf{a}_{t+1}^{\text{raw}} | \bar{\mathbf{o}}_{t+1}), \\ y_{t,k} &= r_t + \gamma(1 - d_t) [z'_{(k)}(\bar{\mathbf{o}}_{t+1}, \mathbf{a}_{t+1}^F) - \alpha \ell_{t+1}], \\ \mathcal{Y}_t &= \text{sg}(\{y_{t,k}\}_{k=1}^K), \quad K < 2N_q. \end{aligned} \quad (14)$$

Thus target critics and replay share the same projected-action support, while entropy remains the density of the raw stochastic policy. Actor and prior updates pass gradients through the non-smooth tire search using

$$\mathbf{a}^{\text{F,ST}} = \mathbf{a}^{\text{B}} + \text{sg}[\mathbf{a}^{\text{F}} - \mathbf{a}^{\text{B}}]. \quad (15)$$

This projected-action surrogate uses the exact projected action forward and an identity backward map through Π_{tire} .

Two quantile heads share $\mathbf{h}_{\zeta} = \mathcal{E}_{\zeta}^Q(\bar{\mathbf{o}}, \mathbf{a}^F)$. For matched $(\bar{\mathbf{o}}_g, \mathbf{a}_g^F, v_g, \mathbf{a}_{\text{th}}^*) \sim \mathcal{D}_g$, let $\mathbf{h}_{\zeta,g} = \mathcal{E}_{\zeta}^Q(\bar{\mathbf{o}}_g, \mathbf{a}_g^F)$; the updates are

$$\begin{aligned} \mathcal{L}_{\text{critic}} &= J_Q + \lambda_g \mathbb{E}_{\mathcal{D}_g} \left[(g_{\psi}(\mathbf{h}_{\zeta,g}) - N_V(v_g))^2 \right], \\ \mathcal{L}_{\text{actor}} &= J_{\pi} + \lambda_p \mathbb{E}_{\mathcal{D}_g} \left[\left\| N_a(\mu_{\theta}^F) - N_a(\mathbf{a}_{\text{th}}^*) \right\|_2^2 \right]. \end{aligned} \quad (16)$$

The offline teacher uses χ^G as privileged rollout information for coupled preview evaluation, whereas the deployed actor and critic are conditioned on $\bar{\mathbf{o}}$ and $(\bar{\mathbf{o}}, \mathbf{a}^F)$, respectively. The matched samples in $\mathcal{D}_g$ therefore distill the teacher action and response-set value into observation-conditioned supervision over the rollout contexts represented in the dataset.

Here J_Q and J_{π} are the standard TQC critic and entropy-regularized actor losses; N_a and N_V normalize actions and teacher values. The actor encoder is independent, whereas the two quantile heads and the training-only robust-value head share the critic encoder. Consequently, $\mathcal{D}$ supplies Bellman transitions, whereas $\mathcal{D}_g$ supplies matched teacher-action and response-set-value supervision. The action prior guides corridor placement, while the auxiliary value labels encourage the shared critic representation to retain response-set robustness information.

3.3. Persistent Tactical Feasible-Region Interface

The state machine carries the selected passing opportunity across successive tactical updates. This temporal persistence complements the spatial corridor: the policy may refine its width and speed request as the encounter evolves, while the committed side remains consistent through the pass. Let Δs_t and Δv_t be the sampled relative quantities defined in Eq. (5). A feasible side is committed as the ego closes on the opponent, retained through SHADOW and OVERTAKE, and released only after a confirmed pass, an abort, or an execution-health failure. Health and abort take precedence over passing, completed passes take precedence over continued overtaking, and an unlocked ego enters DEFEND only when a rear opponent is closing. The pass-confirmation indicator P_t debounces the Pass condition defined after Eq. (5) over N_{post} consecutive tactical updates before the state machine releases the side lock.

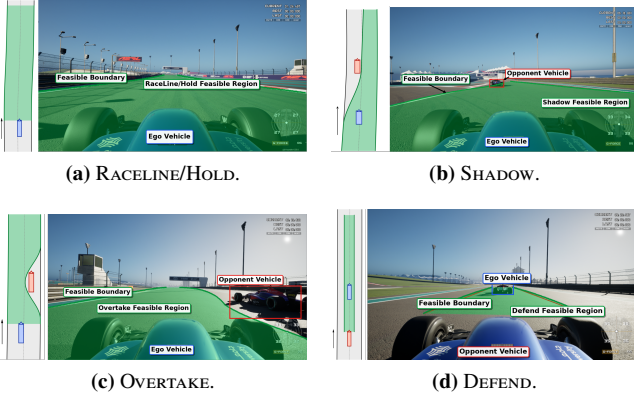


Fig. 2 Illustrative geometry of the persistent tactical modes. Each panel pairs a schematic Frenet representation with the corresponding simulator view and visualizes how the mode shapes the nominal feasible region relative to the ego vehicle, opponent, and track boundaries.

The geometric operation that realizes the selected side is the corridor carver. For $\mathcal{B}_{\text{cap},k} = [\ell_k, u_k]$, the vehicle footprint and one-sided exclusion are

$$\begin{aligned} \rho_{\text{exc},k} &= d_{\text{safe}} + \frac{1}{2} \sum_{j \in \{e,o\}} (L_j |\sin \xi_{j,k}| + W_j |\cos \xi_{j,k}|), \\ \mathcal{B}_{\text{cap},k} &= [\max\{n_{\text{trk},k}^-, -R^{\text{cap}}\}, \min\{n_{\text{trk},k}^+, L^{\text{cap}}\}], \\ C_k^{\text{side}}(\sigma, \eta) &= \begin{cases} [\ell_k + \eta[n_{o,k} + \rho_{\text{exc},k} - \ell_k]_+, u_k], & \sigma = +1, \\ [\ell_k, u_k - \eta[u_k - n_{o,k} + \rho_{\text{exc},k}]_+], & \sigma = -1. \end{cases} \end{aligned} \quad (17)$$

Here $\sigma = +1$ and -1 select the left and right passing sides, respectively. With predicted longitudinal gap Δs_k , SHADOW uses $\eta = \text{clip}((g_{\text{chase}} - \Delta s_k)/(g_{\text{chase}} - g_{\text{ot}}), 0, 1)$, whereas OVERTAKE uses full exclusion ($\eta = 1$) on the locked side. RACELINE uses the capped track interval; DEFEND intersects it with a width- w_{safe} interval centered at the clipped offset $n_{o,k} + b_{\text{lat}}$; and HOLD retains only the connected component containing the ego vehicle. Figure 2 illustrates these geometric semantics.

Algorithm 1 summarizes the complete route from offline interaction supervision to online corridor publication. Immediate contraction and smoothed relaxation allow the feasible region to respond quickly to reduced space while expanding progressively as the passing channel opens. The preceding corridor is revalidated for one update when a new interval loses the required width.

Together, the robust Stackelberg response-set teacher, projected-action TQC policy, and persistent tactical feasible-region interface transform opponent-aware interaction reasoning into side-consistent Frenet bounds and speed guidance for constrained motion planning. By adapting the corridor geometry while preserving the selected passing side, the framework maintains a consistent overtaking intention throughout the interaction.

4. Tactical Corridor Execution with Sequential SOCP and RC-NMPC

The execution layer realizes the persistent tactical feasible region as a continuous path-speed reference and actuator commands. The published corridor defines the spatial optimization domain, while the speed request sets the tactical pace throughout the maneuver.

For the $N = N_{\text{sta}}$ spatial stations, let $\mathbf{z}_k = [n_k, \xi_k]^\top$ and $\mathbf{z}_T = [n_T, 0]^\top$, where n_T is the clipped raceline in RACELINE and the terminal corridor midpoint in the other modes. The set $\mathcal{Z}_k =$

Algorithm 1 Game-supervised tactical-corridor policy

Require: teacher states and replay $\mathcal{D}$; online states, track preview, previous $(m, \sigma^{\text{lock}}, C)$, and execution health

- 1: **Offline teacher and training:**
- 2: **for** each teacher state **do**
- 3: combine Sobol samples with left/neutral/right codes; apply $\mathbf{a}^{\text{raw}} \rightarrow \mathbf{a}^{\text{B}} \rightarrow \mathbf{a}^{\text{F}}$
- 4: discard duplicate or infeasible candidates; generate the coupled preview (8) for each candidate pair
- 5: evaluate (9), retain near-best follower responses, and compute (10)
- 6: store every matched $(\bar{\sigma}, \mathbf{a}^{\text{F}}, \mathbf{v}_g, \mathbf{a}_{\text{th}}^*)$ in $\mathcal{D}_g$
- 7: **end for**
- 8: update TQC from $\mathcal{D}$ and the prior/value terms from $\mathcal{D}_g$ using Eq. (16)
- 9: **Online tactical publication:**
- 10: sample $\mathbf{a}^{\text{raw}}$, apply $\Pi_{\mathcal{A}}$, predict the opponent at constant Frenet velocity, and evaluate the available side widths
- 11: **if** h_t is invalid, or a locked state satisfies $\Delta s_t \geq g_{\text{abort}}$ or loses $L_{\text{min}}^{\text{corr}}$ width **then**
- 12: set HOLD and release the side lock
- 13: **else if** the locked ego satisfies the Pass condition defined after Eq. (5) for N_{post} consecutive updates **then**
- 14: set $P_t = 1$, return to RACELINE, and release the lock
- 15: **else if** a lock exists, or a non-neutral feasible side is requested with $0 < \Delta s_t \leq g_{\text{chase}}$ and $\Delta v_t \geq 0$ **then**
- 16: retain/commit that side; select SHADOW for $\Delta s_t > g_{\text{ot}}$ and OVERTAKE otherwise
- 17: **else if** $\sigma^{\text{lock}} = 0$, $-g_{\text{chase}} \leq \Delta s_t < 0$, and $\Delta v_t < 0$ **then**
- 18: select DEFEND
- 19: **else**
- 20: select RACELINE
- 21: **end if**
- 22: construct the mode interval using (17) and the corresponding raceline, defend, or connected-hold rule; use $\eta = 1$ in OVERTAKE and the gap-based ramp in SHADOW
- 23: evaluate (12); if no trial cap is admissible, invoke current-state HOLD and braking fallback
- 24: apply immediate contraction and smoothed relaxation; on width failure, revalidate the preceding corridor for at most one update
- 25: **return** $(m, \sigma^{\text{lock}}, V_{\text{cap}}^{\text{F}}, C)$ if valid; otherwise return braking-fallback status

$\{[n, \xi]^\top : \underline{n}_k \leq n \leq \bar{n}_k, 1 - \kappa(s_k)n \geq \varepsilon_s, |\xi| \leq \xi_{\text{max}}\}$ combines the footprint-eroded corridor and Frenet-domain bounds. Sequential iteration j linearizes Eq. (6) as $\hat{\mathbf{f}}_{s,k}^j = \mathbf{A}_k^j \mathbf{z}_k + \mathbf{B}_k^j \omega_k + \mathbf{b}_k^j$ and solves

$$\begin{aligned} \mathbf{y}^{j+1} &= \arg \min_{\mathbf{y}} \sum_{k=0}^{N-1} w_\omega \sigma_k^\omega + w_T \epsilon_T \\ \text{s.t. } \mathbf{z}_0 &= [n_e, \xi_e]^\top, \\ \mathbf{z}_{k+1} - \mathbf{z}_k &= \frac{\Delta s_{\text{sta}}}{2} (\hat{\mathbf{f}}_{s,k}^j + \hat{\mathbf{f}}_{s,k+1}^j), \\ \mathbf{z}_k &\in \mathcal{Z}_k, \quad |\omega_k| \leq \omega_{\text{max}}, \\ \|[2\omega_k, \sigma_k^\omega - 1]^\top\|_2 &\leq \sigma_k^\omega + 1, \\ |\mathbf{z}_{N-1} - \mathbf{z}_T| &\leq \epsilon_T, \quad \epsilon_T \geq 0. \end{aligned} \quad (18)$$

Here $\mathbf{y} = \{\mathbf{z}_k, \omega_k, \sigma_k^\omega, \epsilon_T\}$, $\epsilon_T = [\epsilon_n, \epsilon_\xi]^\top$, $\mathbf{w}_T = [w_{n,T}, w_{\xi,T}]^\top$, and $\omega_{\text{max}} = \tan \delta_{\text{max}} / (l_f + l_r)$. The dynamics apply for $k = 0, \dots, N-2$ and the node constraints for $k = 0, \dots, N-1$. The conic epigraph penalizes squared path curvature, while ϵ_T softens the terminal target. Each sequential update is re-evaluated with the nonlinear Frenet model.

The forward-backward sweep converts the tactical speed request into a tire-envelope-screened speed reference over the optimized path. Let $\Delta \ell_k = \|\mathbf{p}_{k+1} - \mathbf{p}_k\|_2$ and $\kappa_{p,k}$ denote the segment length and path curvature. The envelope supplies the traction and

braking limits $a_{T,k}^+$ and $b_{T,k}^-$ at $(\kappa_{p,k}V^2, V)$, after subtracting the resistance acceleration in traction and adding it in braking. The steady tactical request is first converted into a braking-reachable stationwise cap $V_{\text{cap},k}^P$. Intersecting it with the largest envelope-admissible curvature speed gives $V_{\text{lim},k} = \min\{\bar{V}_{\text{curv},k}, V_{\text{cap},k}^P\}$.

The forward pass starts from $V_0^f = V_e$; its terminal value initializes $V_{N-1}^r = \min\{V_{N-1}^f, V_{\text{lim},N-1}\}$. The two recursions are

$$\begin{aligned} V_{k+1}^f &= \text{clip}[V_{\min}, V_{\text{lim},k+1}] \sqrt{(V_k^f)^2 + 2\Delta\ell_k a_{T,k}^+(V_k^f)}, \\ V_k^r &= \min\left\{V_k^f, V_{\text{lim},k}, \sqrt{(V_{k+1}^r)^2 + 2\Delta\ell_k b_{T,k+1}^-(V_{k+1}^r)}\right\}. \end{aligned} \quad (19)$$

Evaluated for $k = 0, \dots, N-2$ and $k = N-2, \dots, 0$, the forward and backward recursions impose acceleration and downstream braking reachability, respectively. Trapezoidal integration of $1/V_k^r$ provides the time stamps used to resample the path-speed reference on the controller grid.

The RC-NMPC tracks the resampled path-speed reference on a fixed prediction grid in an ego-centered local frame. Its state and input are $\chi_k = [x_k, y_k, \theta_k]^\top$ and $\nu_k = [V_k, \delta_k]^\top$, respectively. The discrete kinematic bicycle model is

$$f_{\text{kb}}(\chi_k, \nu_k) = \chi_k + T_c \begin{bmatrix} V_k \cos \theta_k \\ V_k \sin \theta_k \\ V_k \tan \delta_k / (l_f + l_r) \end{bmatrix}. \quad (20a)$$

Here $\nu_k^r = [V_k^r, \delta_k^r]^\top$, with $\delta_k^r = \arctan((l_f + l_r)\kappa_{p,k}^r)$. The quantities $e_{c,k} = (\mathbf{n}_k^r)^\top(\mathbf{p}_k - \mathbf{p}_k^r)$, $e_{\psi,k} = \text{wrap}(\theta_k - \theta_k^r)$ define $\mathbf{e}_k = [e_{c,k}, e_{\psi,k}]^\top$, and $\mathbf{Q} = \text{diag}(q_c, q_\psi)$. The set $\mathcal{U}_{t,k}(\nu_{k-1})$ collects $|\delta_k| \leq \delta_{\max}$, $|V_k - V_{k-1}| \leq \Delta V_{\max}$, and $|\delta_k - \delta_{k-1}| \leq \delta_{\max} T_c$. Its speed component obeys $V_{\min} \leq V_k \leq \min\{v_{x,\max}, V_{\text{cap},t,k}^C\}$, where the stagewise cap is obtained by rate-limiting the tactical request from the previous speed command. Using $\|\mathbf{x}\|_{\mathbf{M}}^2 = \mathbf{x}^\top \mathbf{M} \mathbf{x}$, the controller solves

$$\begin{aligned} \min_{\nu_{0:N_c-1}} \quad & q_{\Delta\delta}(\delta_0 - \delta_{-1})^2 + \sum_{k=0}^{N_c-1} \left(\|e_{k+1}\|_{\mathbf{Q}}^2 + \|\nu_k - \nu_k^r\|_{\mathbf{R}_\nu}^2 \right) \\ \text{s.t.} \quad & \chi_0 = \mathbf{0}, \quad \chi_{k+1} = f_{\text{kb}}(\chi_k, \nu_k), \\ & \nu_k \in \mathcal{U}_{t,k}(\nu_{k-1}), \quad k = 0, \dots, N_c - 1. \end{aligned} \quad (20b)$$

Here $\nu_{-1} = [V_{-1}, \delta_{-1}]^\top$ is the command applied in the preceding control cycle.

Algorithm 2 Sequential SOCP and RC-NMPC Execution

Require: guidance $\mathcal{G}_t$, current state, previous valid reference

- 1: erode the track by the ego footprint, intersect it with C_t , and choose the mode-dependent terminal target
 - 2: initialize the spatial path from the previous valid solution or the corridor midpoint
 - 3: **for** each configured sequential-convexification iteration **do**
 - 4: solve SOCP (18), update the linearization, and evaluate the nonlinear Frenet and corridor residuals
 - 5: stop when the solver and residual tolerances in Appendix A are satisfied
 - 6: **end for**
 - 7: set valid if the path passes the solver, domain, and residual checks
 - 8: **if** valid **then**
 - 9: query $\mathcal{A}_{\text{tire}}(V)$ and construct $V_{\text{cap},k}^P$ from V_{cap}^F , measured speed, and available braking
 - 10: intersect $V_{\text{cap},k}^P$ with the curvature-speed envelope, run (19), and update valid from the tire check
 - 11: **end if**
 - 12: **if** valid **then**
 - 13: time-resample the path-speed reference and rate-limit $V_{\text{cap},t}^F$ to obtain $V_{\text{cap},t,k}^C$
 - 14: solve RC-NMPC (20) and update valid from its status
 - 15: **end if**
 - 16: **if** valid **then**
 - 17: map $(V_{0|t}^*, \delta_{0|t}^{C*})$ to throttle, brake, steering, and gear
 - 18: **else**
 - 19: revalidate the previous reference for one cycle; otherwise request current-state HOLD and issue rate-limited braking with the preceding steering angle
 - 20: **end if**
 - 21: **return** four-channel command and health status
-

5. Experimental results and discussion

5.1. Experimental platform and configuration

The Abu Dhabi Autonomous Racing League (A2RL) provides a full-scale head-to-head racing environment for EAV24 vehicles on Yas Marina North, together with the high-fidelity AutoVerse simulation platform. Using this platform, we established a hardware-in-the-loop (HIL) system that coupled the simulator with the Neousys RGS-8805GC onboard computer used on the EAV24 race car in the full-scale competition (Fig. 3). Both policy training and the in-domain HIL evaluation use Yas Marina North. The AutoVerse simulator ran on a separate simulation server and generated LiDAR and GNSS/IMU data, which were transmitted to the onboard computer over a dedicated Ethernet link using DDSI-RTPS over UDP/IP. All five software modules executed on the RGS-8805GC, which returned throttle, brake, steering, and gear commands to the simulated vehicle through CAN.

The perception, localization, tactical, planning, and control modules operate at 20, 100, 200, 50, and 100 Hz, respectively. The localization module fuses GNSS/IMU measurements to provide the ego vehicle's position, velocity, attitude, and acceleration, while the perception module processes measurements from three LiDAR sensors to estimate the opponent's position, velocity, and heading. Independent zero-mean Gaussian noise is added to the localization outputs with standard deviations of 0.05 m, 0.10 m/s, 0.002 rad, and 0.10 m/s², respectively, and to the perception outputs with standard deviations of 0.10 m, 0.22 m/s, and 0.003 rad, respectively. A 10% independent packet-drop rate is also applied to both output streams before they are supplied to the downstream tactical and planning modules. The tactical module implements the framework presented in Section 3, while the planning and control modules generate the path-speed reference and closed-loop vehicle commands,

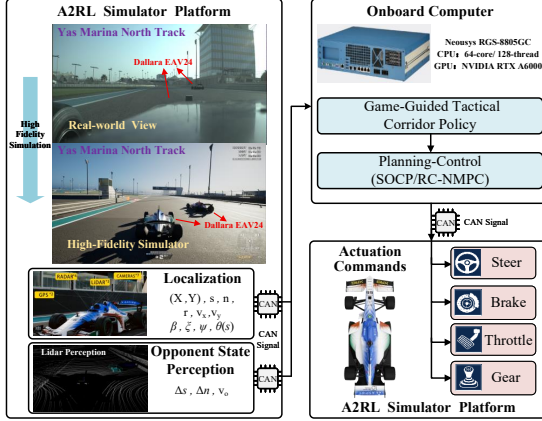


Fig. 3 Hardware-in-the-loop architecture coupling the A2RL high-fidelity AutoVerse simulator with the RGS-8805GC onboard computer used in the full-scale race vehicle.

respectively. Throughout policy training and HIL evaluation, the opponent follows the minimum-curvature racing line.

Statistical results aggregate 1000 Monte Carlo closed-loop laps per setting. In each lap, the opponent initial condition is randomized relative to the ego vehicle in longitudinal position, lateral position, and heading. OSR is the fraction of laps reaching the Pass event defined after Eq. (5), and TTO is the time to that event. All reported mean TTO values are computed over successful laps only. Minimum gap is the smallest net clearance between the ego and opponent body envelopes over a lap. Mean TTC and contact/track-violation rates describe interaction safety, while PFR is the fraction of feasible planning updates.

5.2. Overtaking performance under representative track conditions

To evaluate overtaking performance across different geometries and illustrate the evolution of the learned feasible region throughout each maneuver, Fig. 4 presents four representative encounters at distinct locations on Yas Marina North. The matching labels (b)–(i) link each spatial view of the published corridor and planned path to its synchronized SHADOW or OVERTAKE simulator frame. Across the four encounters, the feasible region evolves from approach guidance into a committed passing channel. In Case 1, the SHADOW corridor at $\Delta s = 18.6$ m places the planned path on the selected side behind the opponent; at $\Delta s = 0.1$ m, the OVERTAKE corridor excludes the opponent footprint and preserves a continuous path along the straight. Case 2 exhibits the same progression through a tight bend: the corridor first establishes the lateral approach at $\Delta s = 17.8$ m and then follows the track curvature while maintaining the locked-side channel at $\Delta s = 0.0$ m. In Case 3, the corridor conforms to the sweeping corner during SHADOW and carries the planned path past the opponent as the relative longitudinal gap decreases from 15.0 to 0.2 m. Case 4 selects the complementary side in the oppositely curved section; its gradual SHADOW offset develops into a fully carved OVERTAKE region as Δs decreases from 14.3 to 0.1 m. These views demonstrate that the state-dependent corridor consistently converts side intention into a connected, track-conforming passing region across straight and curved sections. Cases 1 and 2 are examined in detail as representative high-speed overtakes on a straight and through a consecutive-corner sequence.

Case 1: high-speed overtake on a straight

Case 1 represents a high-speed overtake on the long straight. In Fig. 4, the SHADOW view at $\Delta s = 18.6$ m shows the corridor forming a side-consistent approach channel behind the opponent, with the planned path gradually displaced toward the selected passing side. When the relative longitudinal gap decreases to 0.1 m, the OVERTAKE carver excludes the opponent-occupied region and retains a connected corridor on the locked side.

Figure 5 summarizes the corresponding FSM sequence, Frenet trajectories, and speed response. In panel (b), $s - s_0$ denotes trackwise progress relative to the initial ego position, and n denotes the lateral position relative to the track centerline. The published feasible corridor remains within the physical track boundaries, and the actual ego trajectory stays inside this corridor throughout the overtaking maneuver. Triangles and squares mark the Encounter and Pass events, respectively. Encounter occurs at $\Delta s_E \approx (L_e + L_o)/2$, when the longitudinal body envelopes first overlap after the corridor has established the lateral separation $|\Delta n_E|$. At the Pass marker, the lateral separation satisfies $|\Delta n_P| \geq \rho_{\text{exc},P}$, and the footprint-exclusion condition in Eq. (17) remains satisfied. The actual speed closely follows the desired profile and exceeds 70 m s^{-1} during the pass.

Figure 6 presents the closed-loop realization of this passing channel. The peak absolute lateral contouring, speed-tracking, and heading errors are approximately 0.08 m, 0.35 m s^{-1} , and 0.025 rad, respectively. All recorded tangential-normal acceleration samples remain inside the speed-dependent tire envelope. The committed-side corridor is therefore realized as a trackable path-speed reference throughout the high-speed overtake.

Case 2: high-speed overtake through consecutive corners

Case 2 represents a high-speed overtake through successive curvature changes. In Fig. 4, the SHADOW view at $\Delta s = 17.8$ m shows the corridor following the approaching bend and guiding the planned path toward the selected lateral position. At $\Delta s = 0.0$ m, full OVERTAKE exclusion preserves the same-side channel as the planned path curves alongside the opponent within the track boundaries.

Figure 7 summarizes the corresponding FSM sequence, Frenet trajectories, and speed response. The published feasible corridor follows the changing track boundaries, and the actual ego trajectory remains inside it while traversing the consecutive corners. At Encounter, the curved corridor has already established the lateral separation $|\Delta n_E|$ as the longitudinal body envelopes begin to overlap. At the Pass marker, $|\Delta n_P| \geq \rho_{\text{exc},P}$, preserving lateral clearance throughout the overtaking maneuver. From an entry speed above 60 m s^{-1} , the actual speed closely follows the desired braking and re-acceleration profile throughout the pass.

Figure 8 presents the closed-loop realization of this curved passing channel. The peak absolute lateral contouring, speed-tracking, and heading errors are approximately 0.15 m, 0.35 m s^{-1} , and 0.11 rad, respectively, throughout the successive curvature and speed changes. All recorded tangential-normal acceleration samples remain inside the speed-dependent tire envelope. These results confirm accurate closed-loop realization of the side-locked corridor through consecutive corners with changing track width.

Across 1000 Monte Carlo laps, the full method achieved an OSR of 94.7%, with contact and track-violation rates of 0.0%. The following matched comparisons quantify the performance gains provided by the interactive tactical representation and its constituent modules.

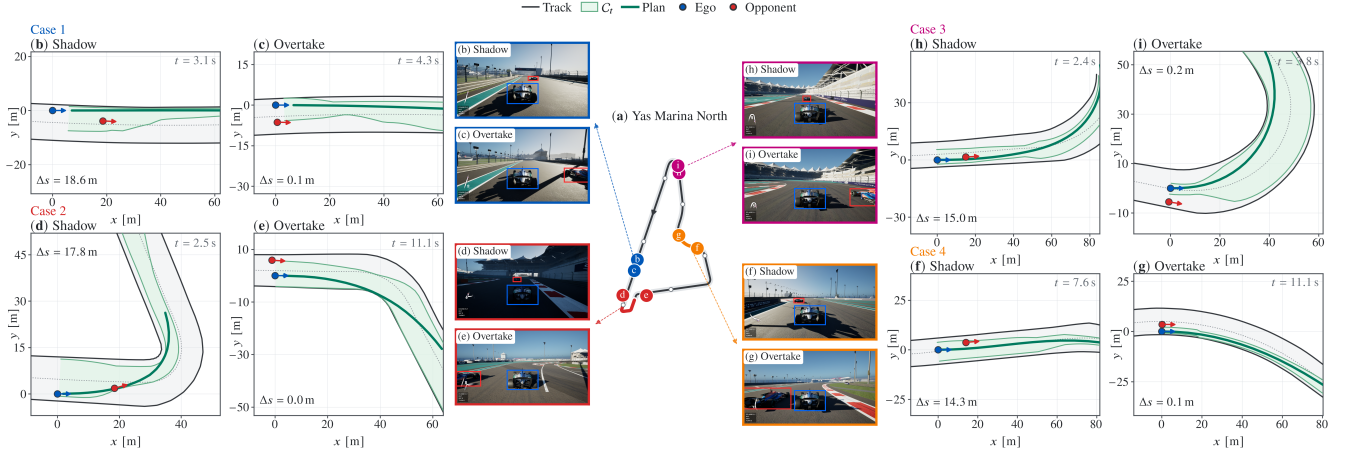


Fig. 4 Representative overtaking situations on Yas Marina North.

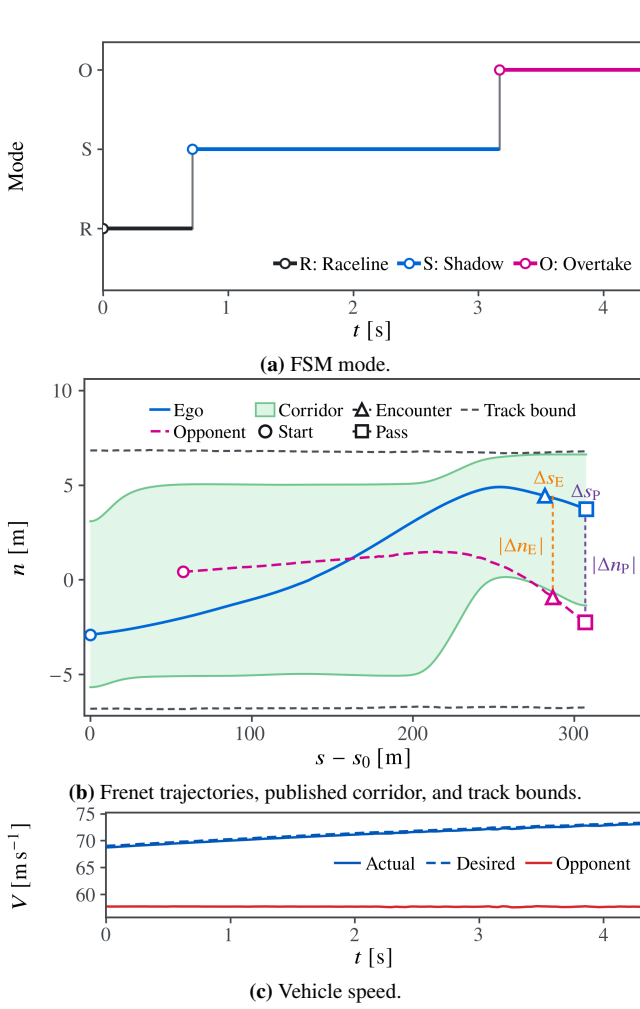


Fig. 5 Tactical-mode sequence, Frenet trajectories and corridor, and speed response in Case 1.

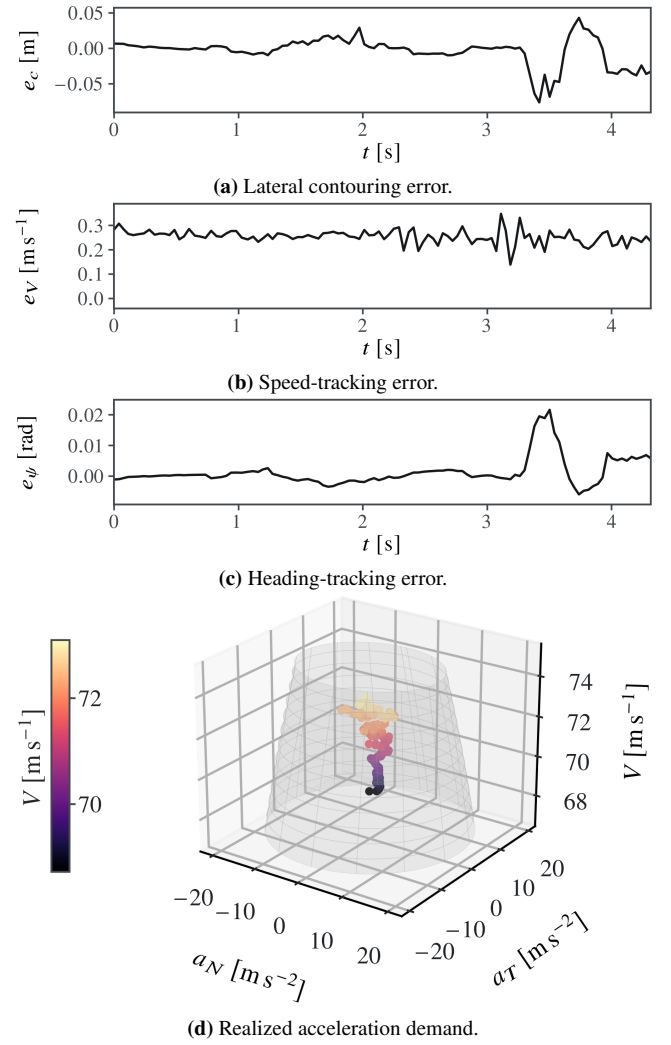


Fig. 6 Closed-loop tracking errors and realized acceleration demand in Case 1.

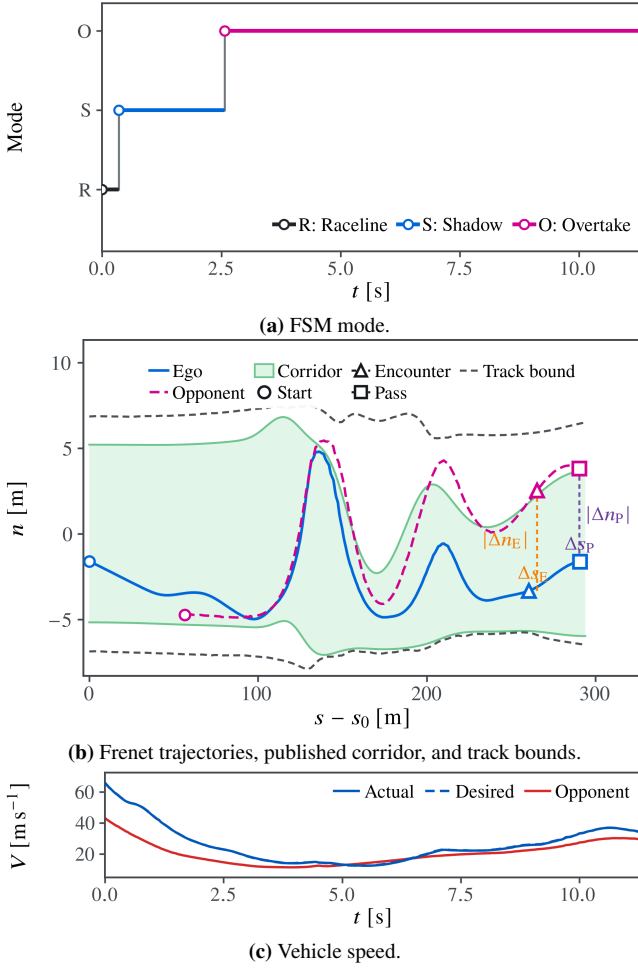


Fig. 7 Tactical-mode sequence, Frenet trajectories and corridor, and speed response in Case 2.

5.3. Comparison with tactical baselines

All methods share the same vehicle plant, race-control interface, corridor projection, path and speed planners, and control module; only the tactical guidance generator differs. *Non-Interactive MPC* predicts the opponent independently of the ego plan and incorporates the predicted occupancy into its obstacle constraints^{33,34}. *Rule-based GT* selects its tactical mode using a fixed-threshold state machine parameterized by $g_{\text{chase}} = 20$ m, $g_{\text{ot}} = 8$ m, and $g_{\text{abort}} = 35$ m. *iLQGames* computes coupled local responses through iterative linear-quadratic game approximations⁹. The outputs of all baselines are mapped to a common tactical interface comprising a passing-side command, a speed cap, and corridor parameters before entering the shared planning and control stack. This setup isolates the effect of tactical guidance on closed-loop overtaking performance. The comparison results are summarized in Table 1, where Prop., NI-MPC, and RB-GT denote the proposed method, Non-Interactive MPC, and Rule-based GT, respectively. Minimum gap is the mean per-lap minimum net clearance between the vehicle body envelopes, and PFR is the fraction of planning cycles with a feasible SOCP solution.

Across 1000 Monte Carlo closed-loop HIL laps per method, the proposed method achieved an overtaking success rate of 94.7% and a mean overtaking time of 4.82 s. Compared with the three baselines, the success rate increased by 31.2–66.2 percentage points and the mean overtaking time decreased by 3.01–6.41 s. Among the

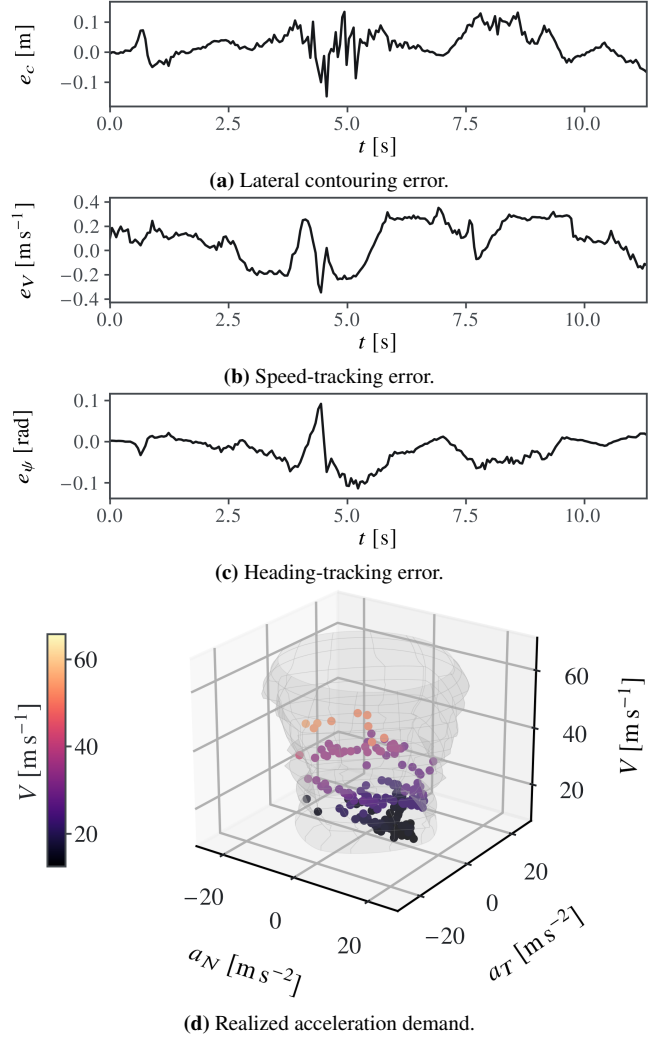


Fig. 8 Closed-loop tracking errors and realized acceleration demand in Case 2.

Table 1 Comparison over 1000 Monte Carlo closed-loop HIL laps per method. Arrows indicate the preferred direction.

Metric	Prop.	NI-MPC	RB-GT	iLQGames
OSR [%] ↑	94.7	28.5	51.6	63.5
TTO [s] ↓	4.82	11.23	9.37	7.83
Min. gap [m] ↑	5.83	3.85	2.14	2.31
Contact rate [%] ↓	0.0	0.7	4.8	1.4
Mean TTC [s] ↑	8.52	6.92	5.64	5.73
Track violation [%] ↓	0.0	0.0	0.0	0.0
PFR ↑	1.000	0.998	0.987	0.995

baselines, iLQGames achieved the highest success rate of 63.5%, demonstrating the benefit of coupled interaction reasoning. The further improvement obtained by the proposed method demonstrates the advantage of expressing tactical reasoning as a persistent feasible region for closed-loop execution. Compared with iLQGames, the proposed method increased the mean minimum gap from 2.31 to 5.83 m and reduced the contact rate from 1.4% to zero. A PFR of 1.000 indicates that the common SOCP planner found a feasible solution at every planning cycle. All 53 unsuccessful laps of the proposed method reached the end of the evaluation lap without a Pass event; none involved contact, a track-boundary violation, or an unrecovered execution-health fault.

5.4. Ablation study

Four ablation variants evaluate the contributions of robust response-set supervision, hysteretic side commitment, adaptive corridor and speed guidance, and TQC-based distributional value learning while retaining the same downstream planning and control stack. *Without game guidance (No-GG)* removes the response-set-optimal action prior and action-conditioned robust-value labels. *Without FSM (No-FSM)* removes hysteretic mode transitions and side locking, allowing the passing-side command to follow the current policy output at each tactical update. In the *Fixed corridor and speed guidance (Fixed-C/S)* variant, the learned corridor-width, clearance, lateral-bias, and speed-cap outputs are replaced with fixed nominal values. The final variant replaces TQC with Soft Actor-Critic (SAC) under the same observation and action spaces, actor architecture, reward function, replay protocol, and training budget.

Table 2 Component ablations over 1000 Monte Carlo closed-loop HIL laps per variant. Arrows indicate the preferred direction.

Metric	Full	No-GG	No-FSM	Fixed-C/S	SAC
OSR [%] ↑	94.7	85.4	82.3	76.1	88.5
TTO [s] ↓	4.82	6.38	6.71	8.94	5.53
Contact rate [%] ↓	0.0	0.9	1.2	3.1	0.5
Min. gap [m] ↑	5.83	2.86	2.17	1.05	3.94
Mean TTC [s] ↑	8.52	4.93	4.31	2.86	6.17
PFR ↑	1.000	0.983	0.974	0.961	0.992

The full method achieved the best performance across all reported metrics, including an OSR of 94.7%, a mean TTO of 4.82 s, a zero contact rate, and a PFR of 1.000 (Table 2). Adaptive corridor and speed guidance provided the largest performance gain. Compared with Fixed-C/S, the full method increased OSR by 18.6 percentage points, reduced the contact rate from 3.1% to zero, and increased PFR from 0.961 to 1.000. Hysteretic mode transitions and persistent side commitment increased OSR by 12.4 percentage points over No-FSM and reduced the contact rate from 1.2% to zero, maintaining a consistent passing intention across successive replanning updates. Robust response-set supervision increased OSR by 9.3 percentage points, reduced TTO by 1.56 s, and increased the mean minimum gap from 2.86 to 5.83 m relative to No-GG, demonstrating its contribution to overtaking success, efficiency, and clearance. TQC increased OSR by 6.2 percentage points over SAC and reduced the contact rate from 0.5% to zero, highlighting the advantage of TQC-based distributional value learning for robust overtaking decisions.

5.5. Hyperparameter sensitivity analysis

Each hyperparameter setting was independently trained and evaluated over 1000 Monte Carlo closed-loop HIL laps under the same

experimental conditions. The nominal setting uses $\varepsilon_{BR} = 0.05$ and $\lambda_P = \lambda_g = 1$. For each value of ε_{BR} , the corresponding teacher labels were regenerated before training. The nominal setting provided the strongest overall balance, achieving the highest OSR, a zero contact rate, and full planning feasibility (Table 3). Reducing ε_{BR} from 0.05 to 0.025 shortened TTO from 4.82 to 4.71 s, while OSR decreased to 92.4% and the minimum gap decreased from 5.83 to 4.55 m. Increasing ε_{BR} to 0.10 enlarged the minimum gap to 6.22 m, while TTO increased to 5.70 s. These results show that ε_{BR} regulates the balance between overtaking efficiency and spatial clearance. Setting λ_P or λ_g to zero reduced OSR to 90.8% and 92.3%, respectively, confirming the contribution of both the action-prior and robust-value supervision signals. Doubling either loss weight maintained OSR above 93% and PFR at or above 0.9968, demonstrating stable performance under increased supervision weights.

Table 3 Hyperparameter sensitivity over 1000 Monte Carlo closed-loop HIL laps per setting. Arrows indicate the preferred direction.

Setting	OSR [%] ↑	TTO [s] ↓	Min. gap [m] ↑	Contact rate [%] ↓	PFR ↑
Nominal	94.7	4.82	5.83	0.0	1.0000
$\varepsilon_{BR} = 0.025$	92.4	4.71	4.55	0.6	0.9926
$\varepsilon_{BR} = 0.10$	93.0	5.70	6.22	0.1	0.9980
$\lambda_P = 0$	90.8	5.70	4.28	0.7	0.9916
$\lambda_P = 2$	93.1	5.29	5.51	0.3	0.9968
$\lambda_g = 0$	92.3	5.31	4.85	0.5	0.9938
$\lambda_g = 2$	93.4	5.21	5.78	0.1	0.9980

5.6. Evaluation on unseen circuits

The tactical policy trained on Yas Marina North was evaluated on the unseen Imola and Suzuka circuits without retraining or parameter retuning. All planning and control parameters were also kept unchanged.

Figures 9 and 10 each present four representative overtaking events. Every event pairs the SHADOW and OVERTAKE phases and localizes them on the corresponding track map. Imola features curved approaches and varied corner geometries, while Suzuka combines rapid direction changes, tight turns, and long acceleration sections. Together, these cases evaluate whether the learned tactical feasible-set representation sustains consistent overtaking decisions across new circuit geometries.

Across the eight illustrated events, the paired simulator views show the tactical state progressing from SHADOW to OVERTAKE on both straight and curved sections. Without retraining or parameter retuning, the framework achieved OSRs of 91.8% and 88.9% on Imola and Suzuka, respectively, while maintaining a zero contact rate on both circuits. The mean minimum gap remained above

Table 4 Aggregate performance over 1000 Monte Carlo closed-loop HIL laps per unseen circuit.

Metric	Imola	Suzuka
OSR [%]	91.8	88.9
TTO [s]	5.19	4.08
Min. gap [m]	5.46	5.05
Mean TTC [s]	8.07	7.45
Contact rate [%]	0.0	0.0
Track-violation rate [%]	0.0	0.1
PFR	0.9968	0.9934

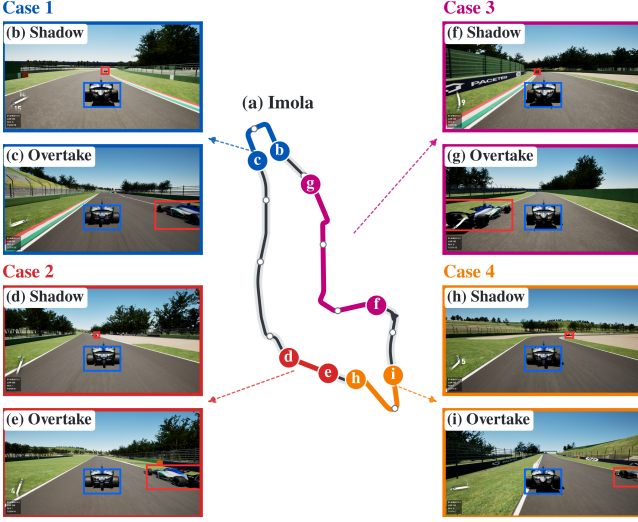


Fig. 9 Representative overtaking events on the unseen Imola circuit. The central map localizes the SHADOW and OVERTAKE phases of four events, with the corresponding simulator views arranged around the map. Blue and red boxes indicate the ego and opponent vehicles, respectively.

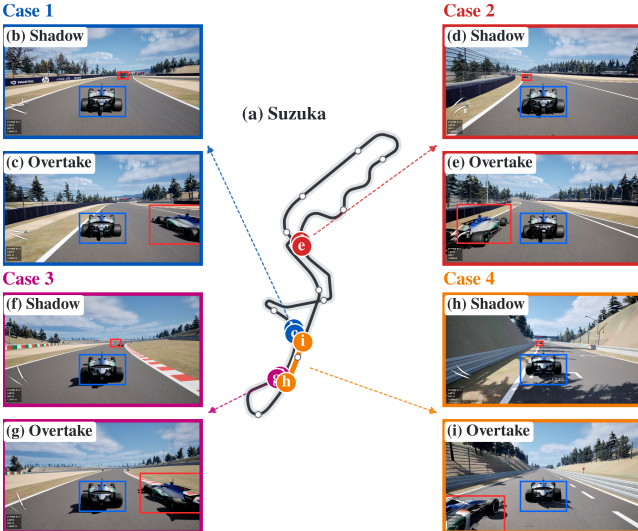


Fig. 10 Representative overtaking events on the unseen Suzuka circuit. The central map localizes the SHADOW and OVERTAKE phases of four events across distinct local geometries. Blue and red boxes indicate the ego and opponent vehicles, respectively.

5.0 m, and PFR remained above 0.993. The track-violation rates were 0.0% on Imola and 0.1% on Suzuka. These results demonstrate effective cross-circuit transfer of the learned tactical feasible-set representation.

6. Conclusion

High-speed overtaking on narrow closed circuits is constrained by limited maneuvering space, increasing the risk of failed passes and collisions. Repeated replanning further complicates the preservation of a consistent passing intention while retaining sufficient geometric freedom for downstream planning and control. To address these challenges, this paper developed a game-guided tactical-corridor decision framework that formulates interactive overtaking as robust tactical feasible-set selection. A robust Stackelberg response-set teacher provides a response-set-optimal action prior and action-conditioned robust-value labels for training a Truncated Quantile Critics (TQC) policy that predicts passing-side intention, corridor parameters, and speed demand. A hysteretic state machine preserves the selected passing side, while deterministic opponent-footprint carving adapts the feasible region to the evolving track and interaction geometry. The resulting side-locked, time-varying Frenet corridor and speed guidance are realized through sequential second-order cone programming (SOCP) planning and reference-conditioned contouring-error nonlinear model predictive control (RC-NMPC).

Monte Carlo closed-loop experiments were conducted on the EAV24 hardware-in-the-loop platform. The proposed framework achieved a 94.7% overtaking success rate, a mean overtaking time of 4.82 s, and a planning-feasibility rate of 1.000, with no collisions or track-boundary violations. Compared with three matched baselines using the same execution stack, the overtaking success rate was 31.2–66.2 percentage points higher and the mean overtaking time was 3.01–6.41 s shorter. Ablation studies further confirmed the roles of adaptive corridor shaping and persistent side commitment. Using the same policy and execution parameters, the framework maintained overtaking success rates of 91.8% and 88.9% on the unseen Imola and Suzuka circuits, respectively. These results demonstrate that persistent tactical feasible regions provide an effective interface between game-guided overtaking decisions and constrained closed-loop execution, highlighting the framework’s practical potential for high-speed autonomous racing.

Future work will leverage participation in the Abu Dhabi Autonomous Racing League (A2RL) to deploy the proposed framework on a full-scale autonomous race car and conduct high-speed on-track validation. The framework will also be extended from the current two-vehicle setting to multi-vehicle racing scenarios to evaluate its performance under more complex competitive interactions.

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References

1. J. Betz, H. Zheng, A. Liniger *et al.*, “Autonomous vehicles on the edge: A survey on autonomous vehicle racing,” *IEEE Open Journal of Intelligent Transportation Systems*, vol. 3, pp. 458–488, 2022.
2. S. Hoffmann, S. Sagmeister, T. Betz *et al.*, “Head-to-head autonomous racing at the limits of handling in the A2RL challenge,” arXiv preprint arXiv:2602.08571, 2026.

3. V. Suresh Babu and M. Behl, "Threading the needle—Overtaking framework for multi-agent autonomous racing," *SAE International Journal of Connected and Automated Vehicles*, vol. 5, no. 1, pp. 33–43, 2022, doi: 10.4271/12-05-01-0004.
4. C. Kim, K. Yi, and J. Park, "Hierarchical motion planning and control algorithm of autonomous racing vehicles for overtaking maneuvers," in *SAE Technical Paper Series*, 2023, Paper 2023-01-0698, doi: 10.4271/2023-01-0698.
5. C. Jung, A. Finazzi, H. Seong *et al.*, "An autonomous system for head-to-head race: Design, implementation and analysis," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 10, pp. 10470–10486, 2023.
6. R. S. Thakkar, A. S. Samyal, D. Fridovich-Keil, Z. Xu, and U. Topcu, "Hierarchical control for head-to-head autonomous racing," *Field Robotics*, vol. 4, no. 1, pp. 46–69, 2024, doi: 10.55417/FR.2024002.
7. A. Liniger and J. Lygeros, "A noncooperative game approach to autonomous racing," *IEEE Transactions on Control Systems Technology*, vol. 28, no. 3, pp. 884–897, 2020.
8. M. Wang, Z. Wang, J. Talbot, J. C. Gerdes, and M. Schwager, "Game-theoretic planning for self-driving cars in multivehicle competitive scenarios," *IEEE Transactions on Robotics*, vol. 37, no. 4, pp. 1313–1325, 2021.
9. D. Fridovich-Keil, E. Ratner, L. Peters, A. D. Dragan, and C. J. Tomlin, "Efficient iterative linear-quadratic approximations for nonlinear multi-player general-sum differential games," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 1475–1481.
10. K. Ji, N. Li, M. Orsag, and K. Han, "Hierarchical and game-theoretic decision-making for connected and automated vehicles in overtaking scenarios," *Transportation Research Part C: Emerging Technologies*, vol. 150, art. 104109, 2023, doi: 10.1016/j.trc.2023.104109.
11. V. Lopez, F. Lewis, M. Liu, Y. Wan, S. Nagesh Rao, and D. Filev, "Game-theoretic lane-changing decision making and payoff learning for autonomous vehicles," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 4, pp. 3609–3620, 2022, doi: 10.1109/TVT.2022.3148972.
12. K. Ji, S. Bae, N. Li, and K. Han, "Competition-aware decision-making approach for mobile robots in racing scenarios," in *Proc. IEEE Intelligent Vehicles Symposium (IV)*, 2024, pp. 878–884, doi: 10.1109/IV55156.2024.10588596.
13. Y. Jia, M. Bhatt, and N. Mehr, "RAPID: Autonomous multi-agent racing using constrained potential dynamic games," in *Proc. European Control Conference (ECC)*, 2023, pp. 1–8.
14. D. Kalaria, C. Maheshwari, and S. Sastry, " α -RACER: Real-time algorithm for game-theoretic motion planning and control in autonomous racing using α -potential function," in *Proc. Learning for Dynamics and Control Conference*, PMLR, vol. 283, pp. 1–18, 2025.
15. Y. Song, H. Lin, E. Kaufmann, P. Dür, and D. Scaramuzza, "Autonomous overtaking in Gran Turismo Sport using curriculum reinforcement learning," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 9403–9409.
16. F. Fuchs, Y. Song, E. Kaufmann, D. Scaramuzza, and P. Dür, "Super-human performance in Gran Turismo Sport using deep reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4257–4264, 2021.
17. P. R. Wurman, S. Barrett, K. Kawamoto *et al.*, "Outracing champion Gran Turismo drivers with deep reinforcement learning," *Nature*, vol. 602, no. 7896, pp. 223–228, 2022.
18. H. Lin, Y. Song, and D. Scaramuzza, "Reaching the limit in autonomous racing: Optimal control versus reinforcement learning," *Science Robotics*, vol. 8, no. 82, art. eadg1462, 2023.
19. B. D. Evans, H. A. Engelbrecht, and H. W. Jordaan, "High-speed autonomous racing using trajectory-aided deep reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 8, no. 9, pp. 5353–5359, 2023, doi: 10.1109/LRA.2023.3295252.
20. M. Hell, G. Hajgató, Á. Bogár-Németh *et al.*, "A LiDAR-based approach to autonomous racing with model-free reinforcement learning," in *Proc. IEEE Intelligent Vehicles Symposium (IV)*, 2024, pp. 258–263, doi: 10.1109/IV55156.2024.10588613.
21. N. Baumann, E. Ghignone, C. Hu, B. Hildisch, T. Hämmerle, A. Bettoni, A. Caron, L. Xie, and M. Magno, "Predictive Spliner: Data-driven overtaking in autonomous racing using opponent trajectory prediction," *IEEE Robotics and Automation Letters*, vol. 10, no. 2, pp. 1816–1823, 2025, doi: 10.1109/LRA.2024.3519878.
22. M. Yuan, J. Shan, and K. Mi, "Deep reinforcement learning based game-theoretic decision-making for autonomous vehicles," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 818–825, 2022, doi: 10.1109/LRA.2021.3134249.
23. X. Hu, Y. Liu, B. Tang, J. Yan, and L. Chen, "Learning dynamic graph for overtaking strategy in autonomous driving," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 11, pp. 11921–11933, 2023, doi: 10.1109/TITS.2023.3287223.
24. S. Han, S. Zhou, J. Wang, L. Pepin, C. Ding, J. Fu, and F. Miao, "A multi-agent reinforcement learning approach for safe and efficient behavior planning of connected autonomous vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 5, pp. 3654–3670, 2024, doi: 10.1109/TITS.2023.3336670.
25. H. Guo, R. Khair, and J. Zhao, "Reinforcement learning from human feedback with fast-slow updates for stable driving strategies," *Control Engineering Practice*, vol. 165, art. 106549, 2025, doi: 10.1016/j.conengprac.2025.106549.
26. R. Reiter, J. Hoffmann, J. Boedecker, and M. Diehl, "A hierarchical approach for strategic motion planning in autonomous racing," in *Proc. European Control Conference (ECC)*, 2023, pp. 1–8, doi: 10.23919/ECC57647.2023.10178143.
27. Y. Liu, A. Zhou, Y. Wang, and S. Peeta, "Proactive longitudinal control to preclude disruptive lane changes of human-driven vehicles in mixed-flow traffic," *Control Engineering Practice*, vol. 136, art. 105522, 2023, doi: 10.1016/j.conengprac.2023.105522.
28. A. J. M. Muzahid, Y. Shi, Z. Wang, A. Zhou, A. Cook, C. R. Wang, and Z. Wang, "Optimizing on-ramp merging for connected and automated vehicles: A hierarchical approach using deep reinforcement learning and optimal control," *Control Engineering Practice*, vol. 165, art. 106614, 2025, doi: 10.1016/j.conengprac.2025.106614.
29. C. Yang, H. Liu, Z. Li, X. Zhang, X. Zhao, and B. Wang, "Hierarchical cooperative planning incorporating MARL guidance and interactive spatio-temporal corridor constraints," *Control Engineering Practice*, vol. 175, art. 107120, 2026, doi: 10.1016/j.conengprac.2026.107120.
30. A. Toschi, F. Prignoli, and M. Bertogna, "Modular decision-making and drivable areas for multi-agent autonomous racing," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 12435–12441, doi: 10.1109/IROS60139.2025.11246897.
31. M. Veneri and M. Massaro, "A free-trajectory quasi-steady-state optimal-control method for minimum lap-time of race vehicles," *Vehicle System Dynamics*, vol. 58, no. 6, pp. 933–954, 2020, doi: 10.1080/00423114.2019.1608364.
32. A. Kuznetsov, P. Shvechikov, A. Grishin, and D. Vetrov, "Controlling overestimation bias with truncated mixture of continuous distributional quantile critics," in *Proc. 37th International Conference on Machine Learning*, PMLR, vol. 119, pp. 5556–5566, 2020.
33. J. Kabzan, M. de la Iglesia Valls, V. J. F. Reijgwart *et al.*, "AMZ Driverless: The full autonomous racing system," *Journal of Field Robotics*, vol. 37, no. 7, pp. 1267–1294, 2020.
34. J. Betz, T. Betz, F. Fent *et al.*, "TUM autonomous motorsport: An autonomous racing software for the Indy Autonomous Challenge," *Journal of Field Robotics*, vol. 40, no. 4, pp. 783–809, 2023.

Appendix A. Algorithm and Evaluation Parameters

This appendix reports the numerical settings and normalized reward features needed to reproduce the algorithmic logic.

Table A.1 Principal model, tactical-policy, training, and closed-loop settings. Symbols follow Eqs. (7) and (13).

Component	Parameter	Value
Vehicle geometry	$m; I_z; l_f, l_r$	742 kg; 1500 kg m ² ;
Tire curves	$F_{z,0}; C_x, C_y$	1.75, 1.35 m 2079.72 N;
Tire curves	$\mu_x, \mu_y; s_x, s_y; q_{x,pk}, q_{y,pk}$	1.6144, 1.5998 2.2345, 1.9745; -0.2555, -0.4787; 0.1296, 0.140
Tire envelope	construction; online evaluation	quasi-steady-state map (3); speed interpolation
Time scales	$T_{tac}; T_C$; station spacing	0.005 s; 0.01 s; 2 m
Tactical bounds	$L^{cap}, R^{cap}; d_{safe}; V_{cap}^{req}$	[1.5, 6.0] m; [0.2, 0.8] m; [15, 90] m s ⁻¹
Gap bounds	$g_{chase}; g_{ot}; g_{abort}$	[18, 24] m; [8, 14] m;
Mode bounds	$b_{lat}; w_{safe}; q_{side}$	[28, 45] m [-1.5, 1.5] m; [1, 5] m; [-1, 1]
Response-set teacher	Sobol candidates per side code; ϵ_{BR}	64; 0.05; 0.99
Projection and FSM	γ_g guard separation Δ_g ; neutral dead band q_{neu} ; N_{post} ; stale-reference limit	0.5 m; 0.25; 3 updates; 1 cycle
TQC	critics; N_q ; retained K ; γ	2; 25; 46; 0.99
TQC guidance	$\lambda_P; \lambda_g$; prior metric	1; 1; identity after action normalization
Training	learning rate; replay capacity; batch size; target-update coefficient	3×10^{-4} ; 10^5 ; 256; 5×10^{-3}
Policy networks	actor encoder; critic outputs; auxiliary head	51 $\rightarrow$ 256 $\rightarrow$ 256; two quantile heads; training only
Reward	w , ordered as in Eq. (13)	(30, -20, -100, -2, -5, 3, -0.1, -0.5, 0.5, 0.25)
Corridor	preview; stations; L_{min}^{corr} ; relaxation factor β	300 m; 150; 3 m; 0.2
Frenet health	$\epsilon_s; \xi_{max}$	0.05; 1.30 rad
Path health	iterations; ($\epsilon_{dyn}, \epsilon_{cor}$)	5; (10^{-3} , 10^{-3} m)
Speed profile	tire-cap grid intervals H_s ; $V_{min}, v_{x,max}$	150; 15, 90 m s ⁻¹
RC-NMPC	steps N_c ; δ_{max} ; $\dot{\delta}_{max}$; ΔV_{max}	15; 0.35 rad; 1.0 rad s ⁻¹ ; 0.5 m s ⁻¹
RC-NMPC weights	$q_{\Delta\delta}, q_c, q_\psi; R_v$	2, 20, 10; diag(1, 0.5)

Here n_{rl} is the raceline offset at s_e , $W_{trk,k} = n_{trk,k}^+ - n_{trk,k}^-$, $W_{trk} = W_{trk,0}$, and $\rho_{exc} = \rho_{exc,0}$. The normalized reward features are

$$\begin{aligned}
 \phi_{ot} &= \mathbb{I}[m_t = \text{OVERTAKE}, m_{t+1} = \text{RACELINE}, P_t], \\
 \phi_{ovb} &= \phi_{ot} \text{ with roles exchanged,} \\
 \phi_{col} &= \mathbb{I}[\text{contact} \vee \text{offtrack} \vee \text{comm_loss}], \\
 \phi_{lat} &= \mathbb{I}[m_t = \text{RACELINE}] \text{clip}\left(\frac{n_e - n_{rl}}{W_{trk}/2}, -1, 1\right)^2, \\
 \phi_{crush} &= \mathbb{I}[|\Delta s_t| < g_{ot}] \left(\frac{[\rho_{exc} - |\Delta n|]_+}{\rho_{exc}}\right)^2, \\
 \phi_{app} &= \mathbb{I}[0 < \Delta s_t < g_{chase}] \text{clip}\left(\frac{\Delta s_{t-1} - \Delta s_t}{g_{chase}}, 0, 1\right), \\
 \phi_{sm} &= \min\left\{1, \frac{\|N_a(a_t^F) - N_a(a_{t-1}^F)\|_2^2}{10}\right\}, \\
 \phi_{gs} &= \min\left\{1, \frac{1}{2N_{sta}} \sum_k \sum_{\pm} \left(\frac{n_{k,t}^{\pm} - n_{k,t-1}^{\pm}}{W_{trk,k}}\right)^2\right\}, \\
 \phi_{sbs} &= \mathbb{I}[|\Delta s_t| < g_{ot}, |\Delta n| \geq \rho_{exc}], \\
 \phi_{wide} &= \frac{1}{N_{sta}} \sum_k \text{clip}\left(\frac{n_{k,t}^+ - n_{k,t}^- - L_{min}^{corr}}{W_{trk,k} - L_{min}^{corr}}, 0, 1\right).
 \end{aligned} \tag{A1}$$